

21-COMPONENT OPERATING, CONNECTION & PIN-MAPPING GUIDE

Reference build for the proposed forest fire / hazard monitoring system

Each component includes a reference picture/illustration, connections, pin/interface mapping, operation procedure, and prior knowledge.

IMPORTANT — VARIANT NOTE

The uploaded project document specifies the 21-component architecture but not exact board/module variants or GPIO assignments. This guide therefore uses clearly labelled reference hardware: classic ESP32 DevKit V1, common DHT11/MQ2/MQ135 modules, NEO-6M, Ai-Thinker Ra-02/SX1278 LoRa, Jetson Nano Developer Kit, I2C LCD1602, LM2596-style buck converter, and a SIM7600-class 4G board. Verify the silkscreen and datasheet of the exact hardware before permanent wiring.

A. MASTER CONNECTION MAP

Sensor node: DHT11 + MQ2 + MQ135 + GPS → ESP32 → LoRa → gateway → Jetson. Camera: IMX219 → Jetson CSI → OpenCV/YOLO. Power: solar panel → charge controller → battery → regulated DC rail → electronics. Backend: Jetson/Python → FastAPI → dashboard/GIS.

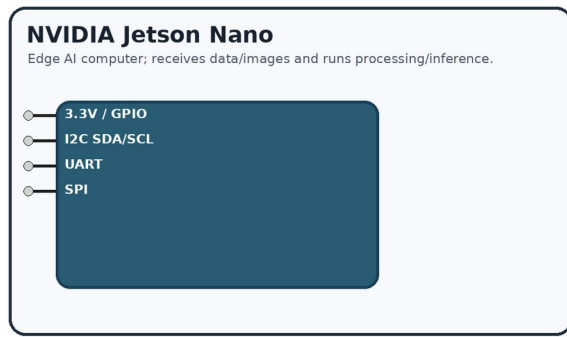
Device	Reference interface / pins	Reference destination
DHT11 module	VCC, DATA, GND	ESP32 3V3, GPIO4, GND
MQ2 module	VCC, GND, AO, DO	ESP32 5V* / GND / GPIO34 via divider / GPIO33 optional via level shift
MQ135 module	VCC, GND, AO, DO	ESP32 5V* / GND / GPIO35 via divider / GPIO32 optional via level shift
NEO-6M	VCC, GND, TX, RX	ESP32 UART2: GPIO16 RX, GPIO17 TX
Ra-02 LoRa	3.3V, GND, SCK, MISO, MOSI, NSS, RST, DIO0	ESP32: 3V3, GND, 18, 19, 23, 27, 25, 26
LCD1602 I2C	VCC, GND, SDA, SCL	ESP32 GPIO21/22 only with voltage-compatible module/level shifter
IMX219	MIPI CSI ribbon	Jetson CSI camera connector
Jetson Nano	40-pin: 3/5 I2C, 8/10 UART, 19/21/23/24 SPI	Peripherals / gateway interface
Solar panel	PV +, PV -	Charge controller PV input
Battery	BAT +, BAT -	Charge controller battery terminals
Buck converter	IN +/-, OUT +/-	Battery/load → regulated rail → electronics
4G module	Power, GND, USB or UART	Jetson or controller; model-specific

*** MQ modules commonly use 5 V heater/power. Their analog outputs may exceed the ESP32 ADC input range; use proper voltage scaling/level shifting and confirm the exact module.**

B. RECOMMENDED BUILD ORDER

- Power/electrical safety → identify exact module models and their voltage requirements.
- ESP32 alone → serial monitor → GPIO/ADC tests.
- DHT11 → temperature/humidity reading.
- MQ2 and MQ135 → raw readings + warm-up + calibration.
- GPS → UART/NMEA data and GPS fix.
- Ra-02 LoRa → two-node transmit/receive test.
- Combine sensor packet → send structured data over LoRa.
- Gateway → receive and forward packets.
- Jetson Nano → Linux/Python and local logging.
- IMX219 → CSI camera test → OpenCV.
- YOLOv8 → offline image/video test → live camera.
- Rules + CNN/SSTM + multi-modal fusion → final hazard confidence.
- FastAPI + GIS dashboard → incidents and sensor markers.
- Solar/battery/enclosure → power and outdoor protection test.
- 10–20 node scaling → range, packet loss, battery and field validation.

1. NVIDIA Jetson Nano



Reference illustration — exact hardware may vary.

Role

Edge computer for local AI/ML, visual processing, hazard inference, Python services, and local logging.

Connection / interface

IMX219 → CSI camera connector. Gateway/peripherals → network/USB or 40-pin interface as applicable. 4G backup can use USB or UART depending module.

Pin map / terminals

Reference Jetson Nano 40-pin: physical 1/17 = 3.3V; 2/4 = 5V; 3 = I2C SDA; 5 = I2C SCL; 8 = UART TX; 10 = UART RX; 19 = SPI MOSI; 21 = SPI MISO; 23 = SPI SCK; 24 = SPI CS0; 26 = SPI CS1; multiple GND pins.

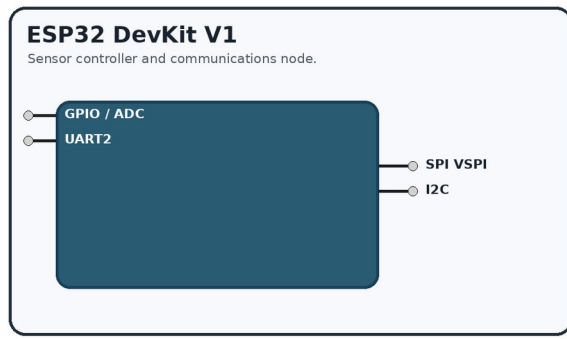
How to operate / test

- Install the supported JetPack/Linux image for your exact Nano variant.
- Boot and verify power, storage, network, camera and 40-pin interfaces.
- Receive/parse sensor packets in Python.
- Run OpenCV/YOLO inference locally.
- Write raw and processed events to local storage.
- Expose processed data through FastAPI.

Prior knowledge: Linux terminal, Python, filesystems, networking, serial interfaces, AI inference.

Critical note: Do not place the board on a conductive surface; use the correct power supply for the exact Nano variant.

2. ESP32 DevKit V1



Reference illustration — exact hardware may vary.

Role

Field controller: reads sensors, labels/structures data, and sends sensor packets over LoRa.

Connection / interface

DHT11 to GPIO4. MQ2/MQ135 analog outputs through safe voltage scaling to GPIO34/35. GPS TX→GPIO16 and RX→GPIO17. Ra-02 SPI to 18/19/23 with CS 27, RESET 25, DIO0 26.

Pin map / terminals

Reference map: GPIO4 = DHT11 DATA; GPIO34 = MQ2 analog input; GPIO35 = MQ135 analog input; GPIO16 = UART2 RX; GPIO17 = UART2 TX; GPIO18 = SPI SCK; GPIO19 = SPI MISO; GPIO23 = SPI MOSI; GPIO27 = LoRa NSS/CS; GPIO25 = LoRa RESET; GPIO26 = LoRa DIO0; GPIO21 = I2C SDA; GPIO22 = I2C SCL; 3V3/GND = power.

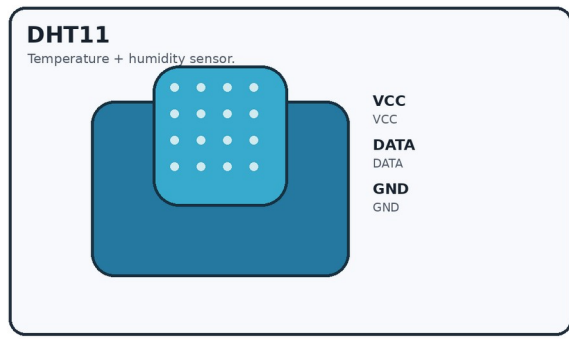
How to operate / test

- Flash a minimal GPIO/Serial program.
- Test each sensor one at a time.
- Read ADCs and record baseline values.
- Parse GPS serial data.
- Initialize LoRa and send a simple HELLO packet.
- Combine readings into structured packets.

Prior knowledge: Digital GPIO, ADC, UART, SPI, I2C, Arduino/ESP-IDF basics.

Critical note: GPIO34/35 are input-only. Keep ESP32 signals within the board's 3.3 V logic limits.

3. DHT11



Reference illustration — exact hardware may vary.

Role

Measures temperature and relative humidity.

Connection / interface

Module VCC→ESP32 3V3; DATA→GPIO4; GND→GND.

Bare sensor versions need the proper pull-up resistor.

Pin map / terminals

3-pin module reference: VCC, DATA, GND. Bare 4-pin sensor: VCC, DATA, NC, GND. DATA is a single-wire digital interface.

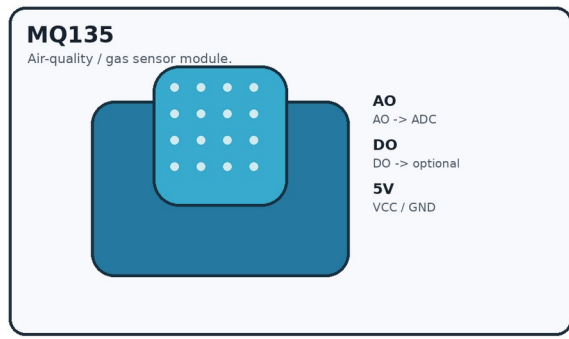
How to operate / test

- Power the sensor.
- Wait for normal startup/stabilization.
- Read no faster than the supported sample rate.
- Check for invalid readings.
- Send temperature and humidity with timestamp/node ID.

Prior knowledge: Digital GPIO and basic sensor libraries.

Critical note: DHT11 is a low-cost, limited-range sensor. Do not use temperature alone as proof of fire.

4. MQ135



Reference illustration — exact hardware may vary.

Role

Provides a gas/air-quality indication for anomaly detection and context.

Connection / interface

Common modules use a 5 V heater/power. AO→ESP32 GPIO35 only through appropriate voltage scaling/conditioning if the output can exceed the ESP32 ADC range. DO is optional and must also be logic-level safe.

Pin map / terminals

Common module: VCC, GND, AO (analog), DO (digital threshold).

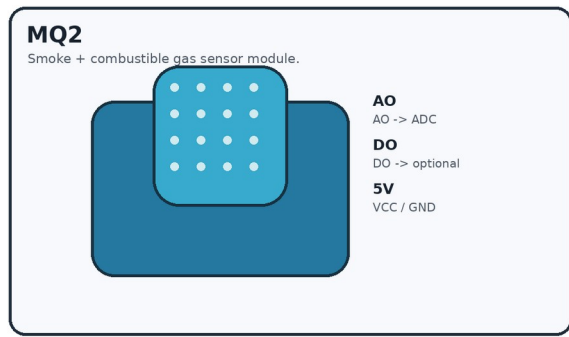
How to operate / test

- Warm-up/burn-in according to the exact sensor/module guidance.
- Measure baseline in known normal air.
- Log AO over time instead of treating a single reading as a diagnosis.
- Calibrate/normalize for your environment.
- Feed the value into thresholds and multi-modal fusion.

Prior knowledge: ADC, voltage dividers/level shifting, calibration, gas-sensor concepts.

Critical note: MQ135 is cross-sensitive and is not a laboratory-grade gas analyzer; do not report exact concentrations without proper calibration.

5. MQ2



Reference illustration — exact hardware may vary.

Role

Detects changes associated with smoke and combustible gases.

Connection / interface

VCC/GND to the appropriate supply; AO→ESP32 ADC through safe scaling; DO optional with level shifting/conditioning.

Pin map / terminals

Common module: VCC, GND, AO, DO.

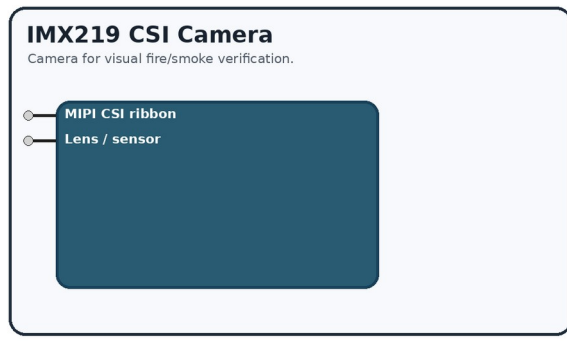
How to operate / test

- Warm up the sensor as required.
- Record a baseline.
- Collect continuous readings rather than relying on one spike.
- Tune an initial threshold from controlled tests.
- Combine MQ2 evidence with temperature and camera evidence.

Prior knowledge: ADC, thresholds, calibration and gas sensor behavior.

Critical note: Do not expose the sensor to dangerous gas concentrations. Use safe controlled test methods.

6. IMX219 CSI Camera



Reference illustration — exact hardware may vary.

Role

Captures visual evidence for fire/smoke detection.

Connection / interface

IMX219 ribbon cable → Jetson Nano CSI camera connector. Verify cable orientation and connector locking before power-up.

Pin map / terminals

No GPIO pin map. It uses a MIPI CSI camera connector and ribbon cable.

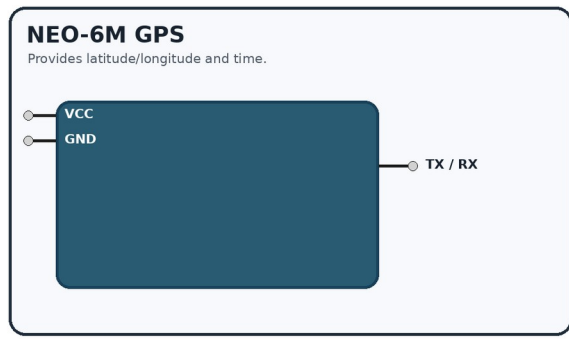
How to operate / test

- Connect the ribbon cable correctly.
- Boot Jetson and confirm the CSI camera is detected.
- Capture a test frame.
- Use OpenCV to read frames.
- Run YOLOv8 on frames and record detection confidence.
- Store incident snapshots when event logic triggers.

Prior knowledge: Linux camera stack, CSI concept, OpenCV, image frames.

Critical note: Do not force the ribbon cable; connector orientation matters.

7. NEO-6M GPS



Reference illustration — exact hardware may vary.

Role

Provides coordinates for incident location tagging and mapping.

Connection / interface

Reference ESP32 UART2: GPS TX→GPIO16 (RX), GPS RX→GPIO17 (TX), plus power and GND. Verify the exact breakout voltage requirements.

Pin map / terminals

Typical module: VCC, GND, TX, RX. UART is the primary interface in common breakouts.

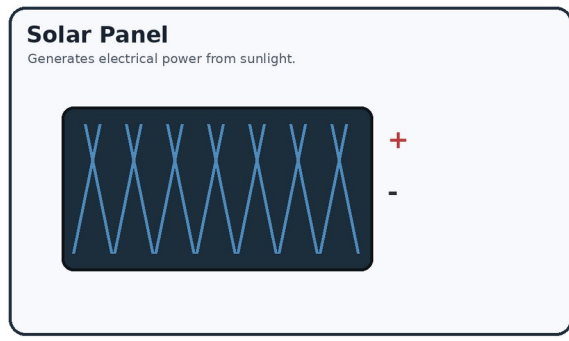
How to operate / test

- Power the module.
- Wait for satellite acquisition outdoors.
- Read NMEA/UBX data at the configured baud rate.
- Validate fix status before accepting coordinates.
- Attach latitude/longitude to every incident packet.

Prior knowledge: UART, TX/RX, NMEA strings, latitude/longitude.

Critical note: GPS accuracy varies with sky visibility and environment.

8. Solar Panel



Reference illustration — exact hardware may vary.

Role

Generates electrical energy for a remote sensor node.

Connection / interface

PV+ / PV- → solar charge controller PV input. Do not connect the panel directly to the battery or ESP32 unless the chosen system is explicitly designed for it.

Pin map / terminals

Power terminals only: PV+ and PV- (or equivalent cable polarity).

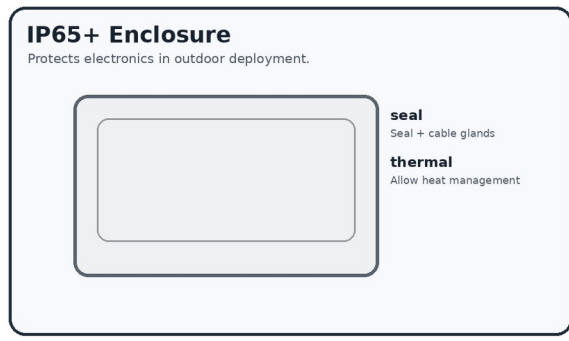
How to operate / test

- Mount for suitable sunlight.
- Check open-circuit voltage/current against the controller specification.
- Route panel output to the charge controller.
- Use weatherproof connectors/cable routing outdoors.

Prior knowledge: Voltage, current, power, polarity, series/parallel basics.

Critical note: Panel voltage can be present whenever illuminated.

9. IP65+ Enclosure



Reference illustration — exact hardware may vary.

Role

Protects electronics against dust and water for outdoor deployment.

Connection / interface

Mount boards inside; route cables through suitable cable glands/connectors; maintain gasket integrity.

Pin map / terminals

No electrical pins. Mechanical/thermal component.

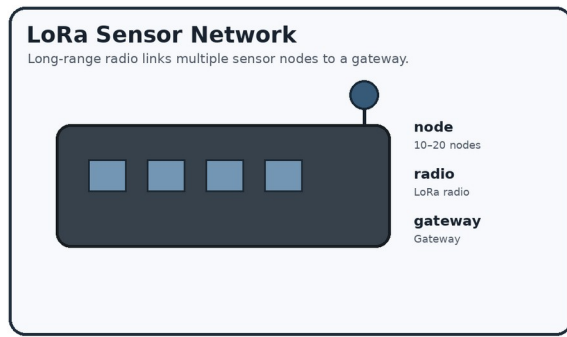
How to operate / test

- Plan cable entry points.
- Mount electronics securely.
- Keep antenna/ventilation requirements in mind.
- Seal unused cable entries.
- Test water/dust protection before field deployment.

Prior knowledge: Mechanical mounting, cable glands, heat management.

Critical note: An enclosure can trap heat; waterproofing does not automatically mean adequate thermal design.

10. LoRa Sensor Network



Reference illustration — exact hardware may vary.

Role

Long-range wireless network joining distributed sensor nodes to a gateway.

Connection / interface

Each node: sensors→ESP32→LoRa radio. Gateway receives packets and forwards them to Jetson/backend. Give every node a unique node_id.

Pin map / terminals

Pin mapping is radio-module specific. For the reference Ra-02 node radio: SPI + CS + RESET + DIO0 + 3.3V + GND.

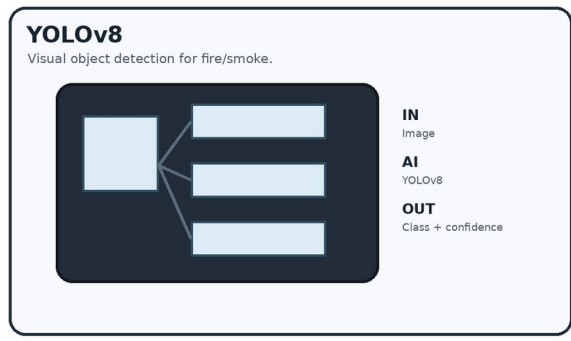
How to operate / test

- Commission one node first.
- Commission one gateway.
- Match frequency/region and radio settings.
- Send known test packets.
- Measure packet loss/RSSI and range.
- Scale gradually to 10–20 nodes.

Prior knowledge: LoRa concepts, radio configuration, packet IDs, gateway/backhaul.

Critical note: Frequency/band and transmitter power must comply with local radio rules and the selected module.

11. YOLOv8



Reference illustration — exact hardware may vary.

Role

Detects visual classes such as fire and smoke from camera frames.

Connection / interface

Camera frame → OpenCV/pre-processing → YOLOv8 model → detected class, bounding box and confidence.

Pin map / terminals

Software only — no hardware pins.

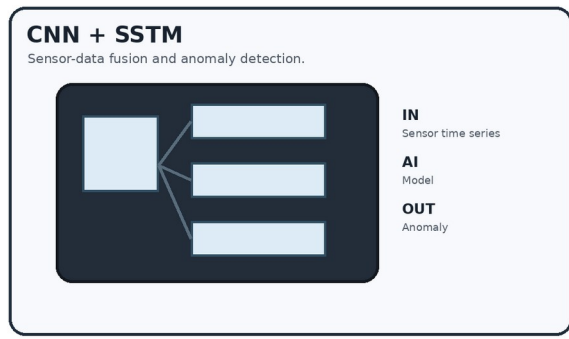
How to operate / test

- Prepare/validate a suitable labeled dataset.
- Start with a known-good pretrained or validated model.
- Run offline images/video before live deployment.
- Set a confidence threshold.
- Log detections with timestamp and sensor-node/location context.
- Use detections as evidence, not the sole final alarm.

Prior knowledge: Python, OpenCV, object detection, datasets and model validation.

Critical note: Lighting, haze, weather and camera placement can cause false positives/negatives.

12. CNN + SSTM



Reference illustration — exact hardware may vary.

Role

Uses sensor sequences for anomaly detection and contextual analysis.

Connection / interface

Structured sensor time-series → preprocessing/features → trained model → anomaly/class score.

Pin map / terminals

Software only.

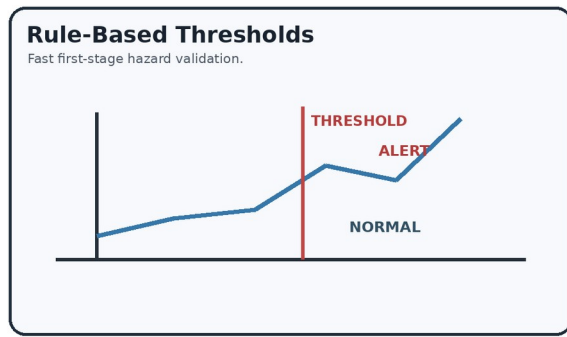
How to operate / test

- Collect normal and abnormal examples.
- Clean/synchronize timestamps.
- Create features/windows.
- Train and validate separately.
- Measure false positives/false negatives.
- Deploy only after field-like validation.

Prior knowledge: Machine learning, time-series data, training/validation, metrics.

Critical note: Do not treat a model score as physically meaningful until validated against real test conditions.

13. Rule-Based Thresholds



Reference illustration — exact hardware may vary.

Role

Fast, interpretable first-stage validation.

Connection / interface

Sensor reading → compare with threshold/range/trend rules → normal/warning/potential incident flag.

Pin map / terminals

Software only.

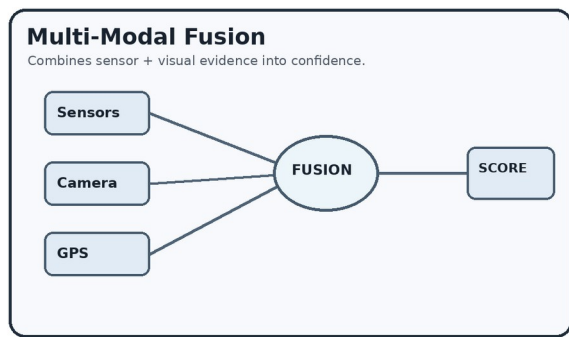
How to operate / test

- Establish baselines.
- Choose initial thresholds from controlled tests.
- Add persistence/time conditions instead of triggering on one sample.
- Log every threshold decision.
- Continuously tune against field data.

Prior knowledge: Conditions, thresholds, hysteresis, rate-of-change concepts.

Critical note: Fixed thresholds can create false positives/negatives, so use them with AI/multi-modal verification.

14. Multi-Modal Fusion



Reference illustration — exact hardware may vary.

Role

Combines independent evidence sources and produces a final confidence/assessment.

Connection / interface

Temperature + humidity + MQ2 + MQ135 + camera + GPS/status → fusion logic/model → confidence + incident class + location.

Pin map / terminals

Software only.

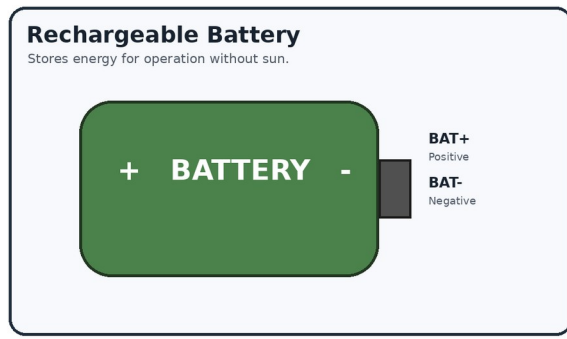
How to operate / test

- Synchronize timestamps.
- Normalize/quality-check sensor streams.
- Create evidence features.
- Combine using transparent rules or a trained fusion model.
- Output confidence, reasons/evidence and location.
- Keep raw data for audit/debugging.

Prior knowledge: Data fusion, confidence scoring, time synchronization, ML basics.

Critical note: Make the final decision traceable: record which sensors/camera contributed.

15. Rechargeable Battery



Reference illustration — exact hardware may vary.

Role

Stores energy so the system can operate when solar generation is low or absent.

Connection / interface

Battery → solar charge controller BAT terminals.
System load should receive power through the intended regulated path.

Pin map / terminals

Power terminals: BAT+ and BAT-.

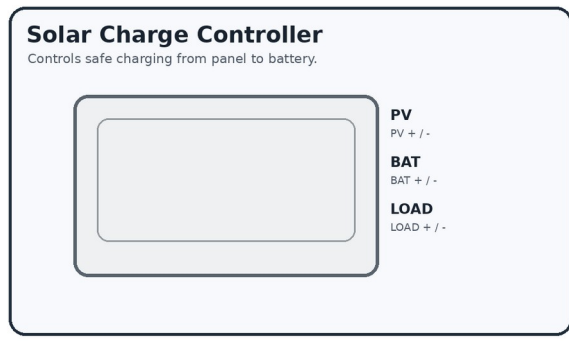
How to operate / test

- Select chemistry/cell count for the controller.
- Estimate daily energy consumption.
- Size capacity for required autonomy.
- Use appropriate protection/BMS where required.
- Monitor voltage and temperature during testing.

Prior knowledge: Battery voltage, Ah, Wh, charging/discharging and BMS concepts.

Critical note: Never use a battery outside its chemistry/charger specifications.

16. Solar Charge Controller



Reference illustration — exact hardware may vary.

Role

Regulates solar charging and protects the battery.

Connection / interface

Solar panel → PV terminals; battery → BAT terminals; regulated system/load path → LOAD output where the controller is designed to support it.

Pin map / terminals

Typical terminals: PV+, PV-, BAT+, BAT-, and possibly LOAD+, LOAD-.

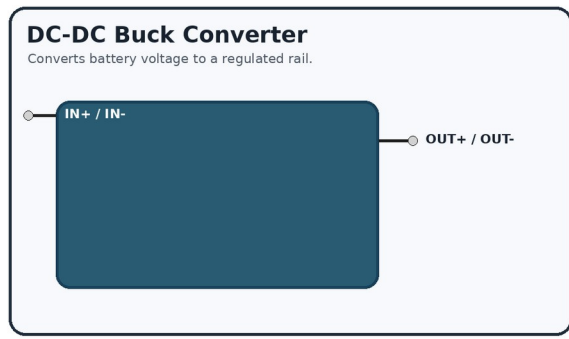
How to operate / test

- Confirm controller battery chemistry/voltage settings.
- Connect battery as instructed by the controller manufacturer.
- Connect solar input.
- Verify charging voltage/current.
- Only then connect loads if the controller provides a rated load output.

Prior knowledge: Charging stages, polarity, battery chemistry, controller ratings.

Critical note: Do not guess terminal order; use the actual controller's markings.

17. DC-DC Buck Converter



Reference illustration — exact hardware may vary.

Role

Creates a stable lower voltage rail from a higher battery/input voltage.

Connection / interface

Battery/controller output → IN+/IN-. Set and measure the desired OUT+/OUT- voltage before connecting electronics.

Pin map / terminals

Typical LM2596-style module: IN+, IN-, OUT+, OUT- plus an adjustment potentiometer.

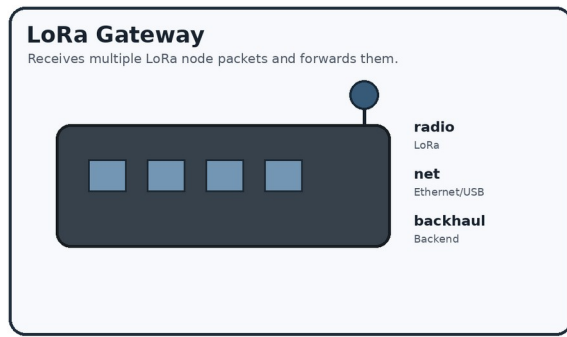
How to operate / test

- Power the module from the intended input.
- Adjust the potentiometer with no sensitive load or a controlled load.
- Measure output with a multimeter.
- Set the exact required voltage.
- Connect the load only after confirming voltage and polarity.

Prior knowledge: Voltage regulation, multimeter use, polarity, current rating.

Critical note: A buck converter does not create energy; ensure input/output current and thermal ratings are adequate.

18. LoRa Gateway



Reference illustration — exact hardware may vary.

Role

Central bridge between multiple LoRa sensor nodes and the backend/Jetson.

Connection / interface

LoRa RF side → distributed sensor nodes.

Host/backhaul side → Jetson/network. Follow the exact gateway manufacturer power/interface requirements.

Pin map / terminals

Gateway hardware is model-specific. Common gateway designs use a LoRa concentrator plus USB/SPI/Ethernet or another host interface. Do not assume the node Ra-02 pinout is the gateway pinout.

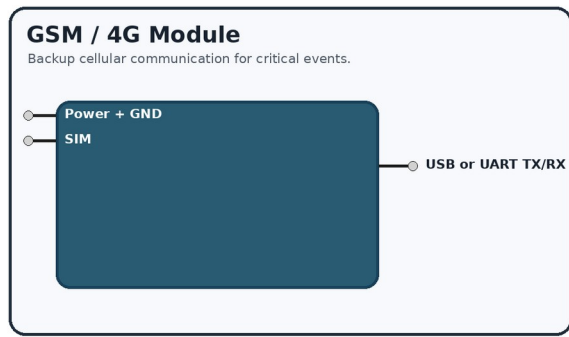
How to operate / test

- Commission gateway hardware.
- Set radio region/frequency configuration.
- Confirm packet reception from one node.
- Forward packets to Jetson/backend.
- Monitor packet rate, duplicates and loss.
- Add more nodes gradually.

Prior knowledge: LoRa gateway architecture, Linux/networking, packet forwarding.

Critical note: Use an appropriate antenna and legal radio configuration.

19. GSM / 4G Module



Reference illustration — exact hardware may vary.

Role

Backup cellular path for critical incident data/alerts.

Connection / interface

Preferred prototype path is often USB to the Jetson/host if the exact board supports it. UART TX/RX can be used when explicitly supported. Insert SIM and connect the required LTE antenna.

Pin map / terminals

Exact pins depend on module. A SIM7600-class board commonly exposes power/GND plus USB and/or UART; many boards also have SIM and antenna connectors.

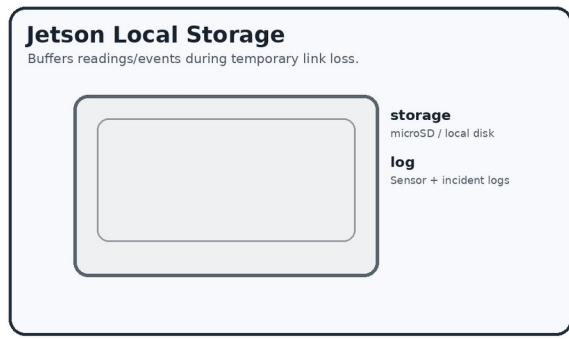
How to operate / test

- Insert activated SIM.
- Attach correct LTE antenna.
- Power from a supply capable of modem current peaks.
- Confirm modem enumeration.
- Configure APN/data connection.
- Send a test request/message before using it as a fallback path.

Prior knowledge: Cellular networking, APN, USB/UART, modem power requirements.

Critical note: Modems can have high current bursts; use the manufacturer's power design.

20. Jetson Local Storage



Reference illustration — exact hardware may vary.

Role

Buffers sensor readings and incident logs locally during temporary communication loss.

Connection / interface

Sensor data → local database/files → retained during network loss → synchronized/uploaded after link restoration.

Pin map / terminals

No GPIO pin. Typical Jetson Nano developer kit uses the microSD slot for main storage.

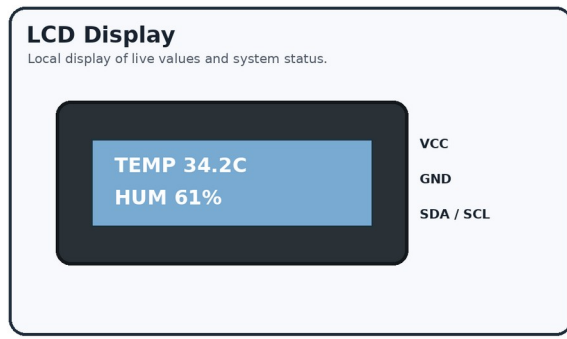
How to operate / test

- Create a structured log/database schema.
- Write every received packet with timestamp/node ID.
- Mark upload state for each record.
- Retry failed uploads.
- Test by disconnecting network and restoring it.
- Verify no events are lost or duplicated.

Prior knowledge: Linux filesystems, SQLite/JSON/CSV or another database, retry queues.

Critical note: Finite storage needs rotation/cleanup and backup strategy.

21. LCD Display



Reference illustration — exact hardware may vary.

Role

Local prototype display for live sensor values and system status.

Connection / interface

Reference ESP32 I2C: SDA GPIO21, SCL GPIO22. Use a 3.3 V-compatible module or level shifting if the backpack pulls lines to 5 V.

Pin map / terminals

Common I2C LCD1602: VCC, GND, SDA, SCL. Parallel LCDs use more GPIO; exact map depends on module.

How to operate / test

- Verify I2C address by scanning.
- Initialize the display.
- Show node ID, temperature, humidity, gas values, LoRa status and GPS fix.
- Use it mainly for local debugging/demo.
- Keep alarm logic independent of the LCD.

Prior knowledge: I2C addressing, text LCD libraries, GPIO/logic levels.

Critical note: Do not expose ESP32/Jetson I2C pins to unsafe voltage.

C. FINAL WIRING & COMMISSIONING CHECKLIST

- Confirm exact part numbers / board variants before wiring.
- Verify VCC and GND on every module.
- Measure the solar/battery/buck power chain with a multimeter before attaching electronics.
- Use a common ground only where the modules are intended to share a ground.
- Never feed 5 V logic directly into an ESP32 3.3 V GPIO/ADC.
- Use voltage dividers/level shifting for MQ sensor outputs where necessary.
- Test each sensor individually before combining the node.
- Test LoRa node-to-node before adding the gateway/backend.
- Test GPS outdoors and validate fix status.
- Test IMX219 independently before enabling YOLO.
- Test rule thresholds using recorded data before live alarm logic.
- Keep raw sensor values, processed values, model outputs, confidence and timestamps.
- Stress-test communication loss and prove local buffering/replay.
- Use unique node IDs for every distributed sensor node.
- Only after bench validation move to the IP65 enclosure and solar/battery field test.

D. SOURCE BASIS

- Uploaded project component-analysis document: the 21-component list and proposed technical stack.
- NVIDIA Jetson Nano developer documentation for 40-pin header, CSI camera, power and microSD/storage interfaces.
- u-blox NEO-6 series documentation for GNSS interfaces.
- Reference Ra-02/SX1278 documentation for LoRa pin and SPI interface.
- Reference ESP32 DevKit V1 pinout for GPIO/ADC/UART/SPI/I2C mapping.
- Exact module datasheets override this guide wherever your purchased hardware differs.